

CE Non-Derivability and the Ultralimit Representation

Investigation:

Current Status

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Abstract

This memorandum records the current state of two related investigations. The first is complete: countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras, and CE non-derivability follows as a corollary. The second, spawned by that proof, asks which purely finitely additive charges on a Boolean algebra arise as pointwise ultralimits of σ -additive probabilities on the same algebra. That problem is settled in every tractable regime — σ -algebras (negative), atomic algebras (positive), algebras embedding join-preservingly into a standard measure algebra (positive), and algebras with too few joins for σ -additivity to bite (positive vacuously) — and the remaining open case is isolated as a single existence question, equivalent by a topological reformulation to a problem about Radon-measure-free compact spaces. The pattern of known results suggests this final question is independent of ZFC; further progress will require forcing or consistency arguments.

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Formal proof: `ce_nonderivability_companion.tex`.

Archival working notes: `finite_cofinite_calculation.md`, `rung3_multidimensional_pfa.md`, `stone_geometric_translation.md`, `row5_candidate.md`, `subalgebra_embedding_lemma.md`.

Status tags used throughout: **Settled** (complete, written up), **Eliminated** (approach shown not to work), **Open** (genuinely unresolved).

1 Main Theorem

Status. Settled. Written up as a companion note targeting *Annals of Pure and Applied Logic*.

In the first-order language $\mathcal{L}_{\text{BA},\mu}$ of Boolean algebras equipped with a normalized finitely additive charge, no first-order theory characterizes those models whose charge extends to a σ -additive measure on the generated σ -algebra. Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras.

The proof is a single ultraproduct argument. Dirac charges δ_n on the finite-cofinite algebra $\mathcal{E}$ over $\mathbb{Q}$ are individually σ -additive. Their ultraproduct along a non-principal ultrafilter $\mathcal{U}$ extending the cofinite filter is the finite-cofinite charge ℓ , which assigns mass 1 to every cofinite set and mass 0 to every finite set. The decreasing sequence $A_k = \mathbb{Q} \setminus \{q_0, \dots, q_k\}$ has empty intersection yet $\ell(A_k) = 1$ for all k , so ℓ is not σ -additive. By Łoś's theorem, any first-order theory satisfied by all δ_n is satisfied by the ultraproduct; but the ultraproduct fails σ -additivity. Contradiction.

Corollary 1 (CE non-derivability). *No accumulation of first-order structural conditions on a directed system of Boolean charge spaces implies collective exhaustion (CE). The finite-cofinite charge witnesses failure of CE while satisfying every first-order condition.*

Remark. The non-axiomatizability theorem relocates a foundational dispute from the domain of terminological disagreement to the domain of logical necessity. The gap between finitely additive coherence and genuine probability cannot be closed by any first-order structural condition: some extra admissibility condition — CE or an equivalent — is logically unavoidable.

2 The Secondary Investigation

The proof above produces the finite-cofinite charge ℓ as an ultraproduct of σ -additive charges on the same algebra $\mathcal{E}$. That raises a sharper fixed-algebra question.

Definition 2 (Ultralimit representability). A finitely additive probability ℓ on a Boolean algebra B is *ultralimit-representable* if there exists a sequence (μ_n) of σ -additive probabilities on B and a non-principal ultrafilter $\mathcal{U}$ on $\mathbb{N}$ such that

$$\ell(A) = \lim_{\mathcal{U}} \mu_n(A) \quad \text{for all } A \in B.$$

The question is: for which Boolean algebras B is every finitely additive probability on B ultralimit-representable?

This has a clean global dichotomy: on a σ -algebra the answer is never (the Nikodym obstruction), on a general Boolean algebra it depends on the algebra's structure. The investigation below classifies the answer by regime.

3 What Is Settled

3.1 The σ -algebra case

Status. Settled (negative).

Theorem 3 (Nikodym obstruction). *Let Σ be a σ -algebra. No purely finitely additive charge on Σ is ultralimit-representable.*

Proof sketch. Any pointwise ultralimit of σ -additive probabilities on a σ -algebra is itself σ -additive, by the Nikodym–Vitali–Hahn–Saks convergence theorem [1], IV.9.8. The non-trivial step is the interchange of the ultralimit with the countable sum $\sum_k \mu_n(A_k)$; it does not follow from the pointwise identity alone. $\square$

Consequence. The representation question is entirely about Boolean algebras that are not σ -complete. The finite-cofinite algebra is the minimal natural setting where the Nikodym obstruction disappears.

3.2 The atomic case

Status. **Settled** (positive, uniformly).

If B is atomic, every σ -additive probability on B is a countable convex combination of Dirac masses at atoms. Atoms of B correspond to principal ultrafilters $\text{Prin}(B) \subseteq \text{St}(B)$, and atomicity of B is equivalent to $\text{Prin}(B)$ being dense in $\text{St}(B)$. This density implies that every finitely additive probability on B is a pointwise limit of σ -additive ones.

Theorem 4 (Atomic case). *If B is atomic, every finitely additive probability on B is ultralimit-representable.*

This covers: the finite-cofinite algebra $\mathcal{E}$, all partition extensions $\mathcal{E}_k$ (generated by $\mathcal{E}$ together with a finite k -partition of $\mathbb{N}$ into infinite pieces), and $\mathcal{E}_\omega$ (countably many pieces). In each case $\text{pfa}(\mathcal{E}_k) \cong \Delta_{k-1}$, and the "spread mass uniformly within each piece" construction gives the approximating sequence explicitly.

3.3 The concrete non-atomic case

Status. **Settled** (positive).

Lemma 5 (Subalgebra embedding lemma). *Suppose B embeds join-preservingly into a σ -algebra Σ carrying a strictly positive σ -additive probability. Then $\text{Supp}_\sigma(B) = \text{St}(B)$: every ultrafilter on B lies in the support of some σ -additive probability on B . In particular, every finitely additive probability on B is ultralimit-representable.*

Here *join-preserving* means: whenever $\bigvee_n a_n$ exists in B , its image in Σ equals $\bigcup_n \iota(a_n)$. Restriction of the ambient σ -additive probability to B then directly inherits σ -additivity.

This covers all concrete non- σ -complete non-atomic algebras: interval algebras, Borel subalgebras of standard measure spaces, and the clopen algebra of any compact metrizable space carrying a strictly positive Borel measure.

Corollary 6. *Every countably generated non-atomic Boolean algebra is ultralimit-representable.*

Proof sketch. A countably generated Stone space is compact metrizable. If B is also non-atomic, the Stone space has no isolated points. By Brouwer's theorem, a compact metrizable zero-dimensional space with no isolated points is homeomorphic to the Cantor set. The homeomorphism provides the join-preserving embedding into the Borel σ -algebra of the Cantor set, and Lebesgue measure is strictly positive on all clopens. $\square$

3.4 The vacuous case

Status. **Settled** (positive, vacuously).

When B has very few countable joins, σ -additivity becomes vacuous: the condition $\mu(\bigvee_n a_n) = \sum_n \mu(a_n)$ only fires when $\bigvee_n a_n$ exists in B .

Theorem 7. *Every finitely additive probability on $\mathcal{Q} = \mathcal{P}(\mathbb{N})/\text{fin}$ is σ -additive on $\mathcal{Q}$.*

Proof sketch. A pairwise disjoint sequence (A_n) in $\mathcal{Q}$ with infinitely many nonzero terms has no join: if $[S]$ were the join, then for each n the selector element $a_n \in A_n \cap S$ yields a strictly smaller upper bound $[S \setminus \{a_n\}]$, contradicting minimality. The only disjoint sequences with an existing join have at most finitely many nonzero terms, for which σ -additivity reduces to finite additivity. $\square$

3.5 The canonical decomposition theorem

Status. **Settled.**

Theorem 8 (Canonical decomposition). *For any Boolean algebra B and any element $A \in B$,*

$$B \cong (B \restriction A) \times (B \restriction A^c)$$

canonically, where $B \restriction A = \{C \in B : C \leq A\}$. The isomorphism sends $C \mapsto (C \wedge A, C \wedge A^c)$.

If ν is σ -additive on $B \restriction A$, then $\tilde{\nu}(C) = \nu(C \wedge A)$ is σ -additive on B : meets distribute over existing joins in any Boolean algebra, so $(\bigvee_n C_n) \wedge A = \bigvee_n (C_n \wedge A)$ exists in $B \restriction A$ whenever $\bigvee_n C_n$ exists in B , and σ -additivity transfers directly.

Corollary 9 (Failures are direct-product failures). *If $\text{Supp}_\sigma(B) \subsetneq \text{St}(B)$, then there exists a nonzero $A \in B$ such that $B \restriction A$ is measure-free and $B \cong (B \restriction A) \times (B \restriction A^c)$.*

Proof. Take any A whose clopen $\hat{A} \subseteq \text{St}(B)$ is disjoint from $\text{Supp}_\sigma(B)$. Every σ -additive probability on B assigns $\mu(A) = 0$. If $B \restriction A$ admitted a σ -additive probability ν , the extension $\tilde{\nu}$ would satisfy $\tilde{\nu}(A) = \nu(A) = 1$, contradicting $\mu(A) = 0$ for all such μ . $\square$

There is no such thing as an *indecomposable* failure of full σ -additive support: any failure is witnessed by a direct-product decomposition.

3.6 The weak distributivity thread

Status. **Settled** (does not apply to the non- σ -complete regime).

Džamonja–Plebanek [2] prove in ZFC: for a Boolean σ -algebra, weakly distributive plus strictly positive finitely additive measure is equivalent to strictly positive σ -additive measure. The same result appears in Fremlin [4], §391D, and Plebanek [6].

These theorems are sharp ZFC results in the σ -complete case. None applies to non- σ -complete algebras: the proofs use countable suprema in essential ways, and σ -additivity on a non- σ -complete algebra is already weaker (it fires only when joins exist in B). Weak distributivity yields no necessary condition on a Strategy D counterexample (see Section 5).

4 What Was Ruled Out

Direct-product strategy via $\mathcal{Q}$.

Status. **Eliminated.**

An earlier approach sought a negative example $B = \mathcal{I} \times \mathcal{Q}$ using a claimed fact that $\mathcal{Q} = \mathcal{P}(\mathbb{N})/\text{fin}$ supports no σ -additive probability. That claim is false: every finitely additive probability on $\mathcal{Q}$ is σ -additive (§3.4). The strategy is eliminated.

Indecomposable negative instances.

Status. **Eliminated.**

An "indecomposable" negative instance would require $\text{Supp}_\sigma(B) \subsetneq \text{St}(B)$ without any direct-product decomposition. The canonical decomposition theorem shows this is impossible (§3.5): any gap in σ -additive support forces a direct-product decomposition with a measure-free factor.

Strategies A, B, C.

Status. **Eliminated.**

Three construction strategies — entangled Boolean extensions (A), quotient algebras with measure-free tails (B), and forcing-generic constructions (C) — all required indecomposable negative instances. Since none can exist, all three are eliminated.

Weak distributivity as a non- σ -complete obstruction.

Status. **Eliminated** (scope mismatch).

The Džamonja–Plebanek theorem was identified as a candidate source of a fifth necessary condition. On careful reading it applies only to σ -complete algebras; no analogous result is known for incomplete algebras. The thread is closed.

5 Current Frontier

Strategy D.

Status. **Open.**

Does there exist a non- σ -complete non-atomic Boolean algebra that is measure-free (admits no σ -additive probability)?

If yes: pair it with any Boolean algebra carrying a σ -additive probability to obtain a direct-product negative instance.

If no: any gap in σ -additive support would require a measure-free non- σ -complete non-atomic factor, which would not exist; the open row collapses and the representation problem is positive in all non- σ -complete non-atomic cases.

Four necessary conditions on any counterexample. A Strategy D counterexample must be: (a) non- σ -complete, (b) non-atomic, (c) measure-free, (d) uncountably generated. Conditions (a)–(c) are by definition. For (d): every countably generated non-atomic Boolean algebra carries a strictly positive σ -additive probability, by the Brouwer–Cantor reduction and the sub-algebra embedding lemma (§3.3).

Topological reformulation. By Stone duality, Strategy D is equivalent to:

Does there exist a compact totally disconnected Hausdorff space K with no isolated points, not basically disconnected, and carrying no strictly positive Radon probability measure?

The dictionary is: no isolated points = non-atomicity of $\text{Clop}(K)$; not basically disconnected = $\text{Clop}(K)$ is not a σ -algebra; no strictly positive Radon measure = measure-freeness of $\text{Clop}(K)$. This connects Strategy D to the literature on Radon-measure-free compact spaces.

Set-theoretic sensitivity. The pattern of known results in that literature strongly suggests Strategy D is independent of ZFC.

- Under $\text{MA} + \neg\text{CH}$: compact spaces of weight $< \mathfrak{c}$ carry strictly positive Radon measures [4], §531–534. Strategy D counterexamples, if they exist, would need weight $\geq \mathfrak{c}$ under these axioms.
- Under $\diamond$: measure-free compact spaces of weight ω_1 have been constructed [5, 3]. Whether any such example is totally disconnected, non-atomic, and not basically disconnected remains to be checked.

No ZFC construction of a counterexample is known, and no ZFC proof that none exists.

6 What Would Move the Problem Next

Route 1 — Forcing/consistency. Attempt to construct, under $\diamond$ or a similar principle, a compact totally disconnected non-atomic non-basically-disconnected measure-free space. Entry points: Kunen [5] and Fedorchuk [3] for measure-free compacta; verify the non-atomic and non-basically-disconnected conditions for those constructions. This is a set-theoretic topology project.

Route 2 — Consistency of non-existence. Show, under $\text{MA} + \neg\text{CH}$, that every non- σ -complete non-atomic Boolean algebra carries a strictly positive σ -additive probability. Entry point: Fremlin [4], §531–534. This would establish the positive half of an independence result.

Route 3 — Targeted literature pass. The Radon-measure-free compact space literature may contain results that answer Strategy D directly, or reduce it further. The combined constraint of non-atomicity and non-basic-disconnectedness alongside measure-freeness does not appear to have been studied explicitly. A targeted pass through Fremlin vol. 5 and Plebanek’s survey articles on measures on Boolean algebras is cheaper than a forcing project and might close the gap.

What is not useful. Further ZFC Boolean-algebra arguments. Weak distributivity. The Džamonja–Plebanek theorem and its analogues. These threads are exhausted.

7 Summary

Regime	Answer	Method
σ -complete non-atomic	Negative	Nikodym / Dunford–Schwartz IV.9.8
Atomic	Positive	Density of $\text{Prin}(B)$ in $\text{St}(B)$
Join-preserving into measure algebra	Positive	Subalgebra embedding lemma
Countably generated, non-atomic	Positive	Brouwer + embedding lemma
Too few joins (e.g. $\mathcal{P}(\mathbb{N})/\text{fin}$)	Positive (vacuous)	σ -additivity = finite additivity
Non- σ -complete non-atomic, intermediate joins	Open	Strategy D

Table 1: Classification of the ultralimit representation problem by regime.

The open row requires a measure-free direct-product factor that is itself non- σ -complete and non-atomic. Strategy D asks whether such a factor can exist. The Boolean-algebraic methods employed throughout this investigation do not reach that question; it lives at the set-theoretic and forcing level.

References

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